Gradient descent

Model training

Linear regression

Least squares loss

$$\min_{\vec{\mathbf{w}}} \sum_{(\vec{\mathbf{x}},y)\in \text{train set}} (y - \vec{\mathbf{w}} \cdot \vec{\mathbf{x}})^2$$

How to minimize a function?

Compute derivative, set to 0, and solve.

Quadratic polynomial:

$$ax^2 + bx + c = 0$$

Explicit solution:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Problems?

- Can't always explicitly solve (e.g. degree-5 polynomial).
- Using explicit solution can be time-consuming (Lin. reg.: $\mathcal{O}(N^{2.3})$).

Alternative: **iterative** methods.

Iterative approach

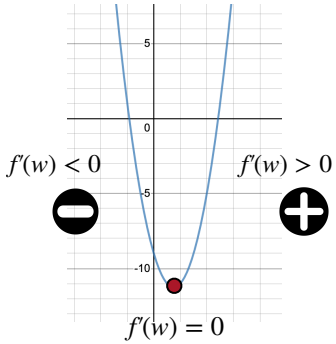
Consider $f(w) = 4w^2 - 6w - 9$.

What do we know about the derivative's behavior, to the left of, right of, and at the minimum?

Idea: Start somewhere on the curve and "walk down" toward its minimum.

If $f'(w) < 0$ (i.e. negative), increase w (i.e. move right).

If $f'(w) > 0$ (i.e. positive), decrease w (i.e. move left).





Iterative approach

Idea: Start somewhere on the curve and "walk down" toward its minimum.

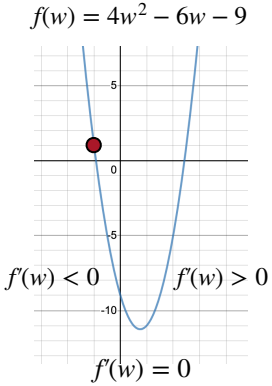
If $f'(w) < 0$, increase w (i.e. move right).

If $f'(w) > 0$, decrease w (i.e. move left).

Fill in table. Initialize $w = -1$ and update it by ± 1 according to above.

iClicker: Write value of w after 3 updates (blue cell). Any problems?

w	$f(w)$
-1	1
0	-9
1	-11
0	-9





Iterative approach

Gradient descent update rule

$w \leftarrow w - \eta \cdot f'(w)$, where
η ("eta"): learning rate

Idea: Update w based on the derivative.

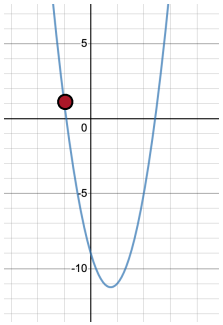
Fill in table. How would we update w?

$f'(w) = 8w - 6$
Let $\eta = 0.1$.

Write value of w after one update.

$f(w) = 4w^2 - 6w - 9$

w	$f(w)$	$f'(w)$
-1	1	-14
0.4	-10.76	-2.8
0.68	-11.23	-0.56



$w \leftarrow w - \eta \cdot f'(w)$
$w \leftarrow -1 - (0.1)(-14)$
$w \leftarrow -1 + 1.4$
$w \leftarrow 0.4$

Multidimensional case: Gradient

Single- vs. multi-variable calculus

Single-variable calculus
one input → one output

$f_1(x) = x^2$
$f_2(x) = \ln(x^3)$
…

Multi-variable calculus
multiple inputs → multiple outputs

$f_3(x, y, z) = x^2y + x^3z^2$

$f_4(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}) = \begin{bmatrix} x_1 + \ln(x_2^3) \\ x_3 - x_1 \end{bmatrix}$



Compute univariate derivative

Single-variable calculus
one input → one output

$f_1(x) = x^2$
$f_2(x) = \ln(x^3)$
…

Try it out: Compute the derivative of f_2.
(Write answer in iClicker poll.)

Answer:

$$\frac{df_1}{dx} = 2x$$

Use chain rule for f_2 (let $u = x^3$):

$$\frac{df_2}{dx} = \frac{df_2}{du}\frac{du}{dx} = \frac{1}{u} \cdot 3x^2$$
$$= \frac{3x^2}{x^3} = \frac{3}{x}$$

Compute partial derivatives

Multi-variable calculus

multiple inputs → multiple outputs

$$f_3(x, y, z) = x^2y + x^3z^2$$

$$f_4\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + \ln(x_2^3) \\ x_3 - x_1 \end{bmatrix}$$

Partial derivative: Derivative of multivar. function w.r.t. a single variable.

$$\frac{\partial f_3}{\partial x}, \frac{\partial f_3}{\partial y}, \frac{\partial f_3}{\partial z}$$

(∂: `\partial` in Latex)

How to compute a partial derivative

1. Treat all other variables as **constants**.
2. Compute the derivative w.r.t. the variable of interest.

$$f_3(x,y,z) = x^2y + x^3z^2$$

$$\frac{\partial f_3}{\partial x} = 2xy + 3x^2z^2$$



Compute partial derivatives

How to compute a partial derivative

1. Treat all other variables as **constants**.
2. Compute the derivative w.r.t. the variable of interest.

$$f_3(x,y,z) = x^2y + x^3z^2$$

$$\frac{\partial f_3}{\partial x} = 2xy + 3x^2z^2$$

Try it out: Compute $\frac{\partial f_3}{\partial y}$.

(Write in iClicker poll.)

Answers:

$$\frac{\partial f_3}{\partial y} = x^2 + 0 = x^2$$

$$\frac{\partial f_3}{\partial z} = 0 + x^3(2z) = 2x^3z$$

Compute gradient

Partial derivatives example

$$f_3(x,y,z) = x^2y + x^3z^2$$

$$\frac{\partial f_3}{\partial x} = 2xy + 3x^2z^2$$

$$\frac{\partial f_3}{\partial y} = x^2 + 0 = x^2$$

$$\frac{\partial f_3}{\partial z} = 0 + x^3(2z) = 2x^3z$$

Gradient: Vector containing the partial derivatives w.r.t. all input variables of a multivariate function.

$$\nabla f_3 = \begin{bmatrix} \frac{\partial f_3}{\partial x} \\ \frac{\partial f_3}{\partial y} \\ \frac{\partial f_3}{\partial z} \end{bmatrix} = \begin{bmatrix} 2xy + 3x^2z^2 \\ x^2 \\ 2x^3z \end{bmatrix}$$

(∇: `\nabla` in Latex)



Poll

Consider the function

$$f(x, y, z) = x^3zy^2 - xz + yz$$

Which of the following is the gradient of f, i.e. $\nabla f = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \\ \partial f/\partial z \end{bmatrix}$?

(a) $\begin{bmatrix} 3x^2y^2 \\ y \\ 3x^2y^2 - x \end{bmatrix}$ (b) $\begin{bmatrix} 3x^2zy^2 - z \\ 2x^3zy + z \\ x^3y^2 - x + y \end{bmatrix}$ (c) $\begin{bmatrix} 2x^3zy + z \\ x^3y^2 + y \\ 3x^2zy^2 - z + y \end{bmatrix}$

Answer: (b).